Lesson notes · for you

Talk in binary

IoT · Level 2 · Lesson 3

Materials
Five small pieces of paper and a pen; Your hand, and a friend to check; Optional: a phone or computer for the binary lab

Learning goals

- Count from 0 to 31 in binary with place values.
- Explain that every reading, message and photo is stored as bits.
- Work out why more bits mean more detail but more data.

Activity: Binary cards

1. Draw 16, 8, 4, 2 and 1 dots on five pieces of paper, and lay them in a row with 16 on the left.
2. Show the cards whose dots add up to your age, and hide the rest. Write 1 under each card that shows and 0 under each hidden one.
3. Do the same for today's date (the day of the month) and for a number up to 31 that a friend picks. Ask the friend to check by adding the dots.
4. Thumb 1, then 2, 4, 8, and 16 on the little finger: a finger up is 1, down is 0. Count from 0 to 31 while a friend checks.
5. In the binary lab, type your name in English and in Arabic. Which takes more bytes, and why?

No kit needed. When you count on your hand, keep your palm facing you, so no finger pattern looks rude to anyone watching.

Discussion

- Why do computers use 1 and 0 instead of the ten digits we use?
- Why does the same message take more data in Arabic than in English?
- When is it worth paying for more bits, and when isn't it?

Check yourself

- Write 22 in binary, and say how many values 6 bits can count.

Try it this week
Binary cards: Make five binary cards: draw 16, 8, 4, 2 and 1 dots on five pieces of paper. Lay them in a row, biggest on the left. Show a card's dots for 1 and hide them for 0, and make your age, today's date (the day of the month), and a number up to 31 that a friend picks. Write each one in binary. Then count from 0 to 31 on one hand, with a friend checking. If you have a phone or computer, type your name in the binary lab in English and in Arabic, and write down how many bytes each takes.

Worksheet

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Name _____ Class _____ Date _____

1 **A small sensor sends each reading using 4 bits. How many different values can it send?**

◯ 15 ◯ 16 ◯ 8

2 **Write 13 in binary, using five bits worth 16, 8, 4, 2 and 1.**

13 = 8 + 4 + _____ , so 13 in five bits is _____ .

Word bank: 1 · 10110 · 01101 · 2

3 **Going from 4 bits to 8 bits doubles the number of values.**

◯ True ◯ False

4 **Match each thing to its rough size.**

One temperature reading	• • • •	Up to 140 bytes
One SMS	• • • •	About 50 million bytes (50 MB)
A page of a book	• • • •	About 3 million bytes (3 MB)
A phone photo	• • • •	A few bytes
A 10-minute lesson video	• • • •	About 2,000 bytes (2 KB)

5 **A light sensor is read with 3 bits (8 levels). A thin cloud passes and the light drops a little, but the reading doesn't change. Why?**

◯ The sensor is broken: a working one shows every change

◯ The change was smaller than one step, so the reading stayed the same

◯ 3 bits can only tell light from dark, nothing in between

✎ **Your phone's data bundle runs out fast. From this lesson, what uses the most data, and what one change would you make?**

Check your answers

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1 **16**

Each bit doubles the count: 2, 4, 8, 16.

2 **13 = 8 + 4 + 1, so 13 in five bits is 01101.**

8 + 4 = 12, and one more makes 13, so the 8, 4 and 1 places are on: 01101.

3 **False**

Each extra bit doubles the count, so four extra bits double it four times: from 16 values to 256, sixteen times as many.

4 **One temperature reading — A few bytes**
One SMS — Up to 140 bytes
A page of a book — About 2,000 bytes (2 KB)
A phone photo — About 3 million bytes (3 MB)
A 10-minute lesson video — About 50 million bytes (50 MB)

The sizes grow from a few bytes to millions of bytes.

5 **The change was smaller than one step, so the reading stayed the same**

With only 8 levels from dark to bright, each step is big, and a small change rounds to the same level.